Graphic Era

Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no. 2556038

Name of the Course: B. Tech.
Semester: II
Name of the Paper: Engineering Chemistry
Paper Code: TCH-201
Time: 1.5 hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Draw the MOT diagram of O_2 molecule. Arrange O_2, O_2^+, O_2^- and O_2^{2-} in increasing order of stability. (CO1)

OR

b. Describe band theory of metallic bond with the help of suitable example. (CO1)

Q2. (10 Marks)

a. Define hydrogen bonding and its conditions, With the help of hydrogen bonding explain why o-nitrophenol can be steam distilled while p-nitrophenol cannot (CO1)

OR

b. Explain the following:
i. Principle of spectroscopy and its advantages
ii. With the help of MOT diagram show the number of non-bonding molecular orbitals in valence shell of HF molecule. (CO1)

Q3. (10 Marks)

a. The hardness of 40,000 liters of sample water was removed by passing it through a zeolite softner. The softner than required 150 litres of NaCl solution containing 100g/l of NaCl. Calculate the hardness of sample water. (CO2)

OR

b. Describe the requirement of lime and soda for removing calcium and magnesium salts hardness. Also give the reactions. What are the disadvantages of lime soda process? (CO2)

Q4. (10 Marks)

a. Explain the Ion-Exchange method of water treatment with the help of diagram. Also discuss the regeneration process and advantages of this method. (CO2)

OR

b. Explain, why hardness of water is expressed in terms of $CaCO_3$ equivalents. A water sample on analysis was found to consist the following impurities-$Mg(HCO_3)_2$ = 32.4 ppm; $Ca(HCO_3)_2$ = 14.6 ppm; $MgSO_4$ = 6.8 ppm; $CaCl_2$ = 9.5 ppm; NaCl = 23.4 ppm. Calculate the temporary and permanent hardness of water. (CO2)

Q5. (10 Marks)

a. Write a short note on the following: i. Scales and Sludges. ii. Caustic embrittlement (CO2)

OR

b. Explain the following:
i. Difference between BMO and ABMO
ii. Nanomaterials with their applications (CO1)